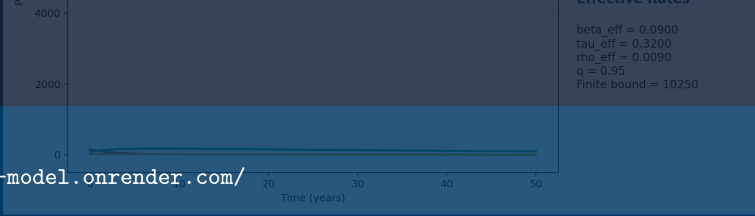
Live dashboard: `https://hiv-model.onrender.com/`



Background / Motivation

Why this topic matters

- HIV/AIDS remains shaped by biological, behavioural, and social factors.
- Rwanda has made strong progress in testing, treatment, and viral suppression, but prevention and adherence remain important.
- Public-health behaviour has memory: awareness, stigma reduction, testing history, and adherence patterns can influence future dynamics.

Presentation focus: the dashboard makes the mathematical model visible, testable, and easier to defend through live simulation.

Rwanda context
HIV prevention

Mathematical lens
Fractional memory

Output
Interactive dashboard

Problem Statement

Gap

- Ordinary HIV models often depend only on the present state.
- Social behaviour is frequently treated as fixed or simplified.
- Many mathematical reports stop at equations and static graphs.

This project responds by

- Formulating a fractional-order SITA model.
- Adding bounded social behaviour interventions.
- Building a working dashboard for simulation, scenario comparison, sensitivity analysis, and exports.

Central question: How do fractional memory and social behaviour interventions affect simulated HIV dynamics in a Rwanda-aware academic dashboard?

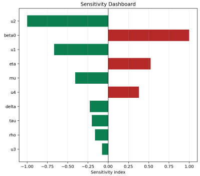
Objectives

Main objective

To formulate, analyse, and computationally simulate a fractional-order SITA model of HIV transmission incorporating social behaviour interventions. **Specific**

objectives

- Construct the SITA model with Caputo fractional derivative.
- Derive the disease-free equilibrium and reproduction number.
- Analyse positivity, boundedness, and threshold behaviour.
- Implement a numerical ABM-type solver and dashboard.
- Compare intervention scenarios, memory effects,

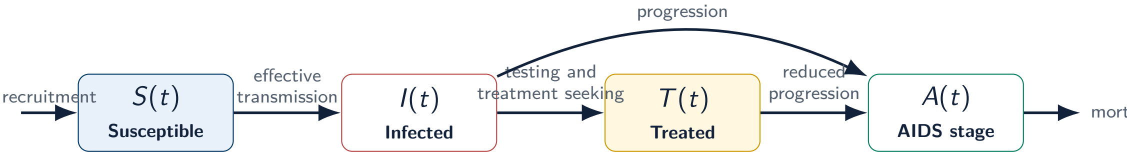


Chapter 6 Result Generator: Baseline Table

Quantity	Value	Interpretation
R0	0.430	Controlled
Peak infected	150.000	at t=0.0 years
Final susceptible	7146.818	S(t_end)
Final infected	3.113	I(t_end)
Final treated	94.272	T(t_end)
Final AIDS	3.588	A(t_end)

Dashboard evidence is the central presentation artifact.

Model Diagram: SITA



Intervention effects: u_1 awareness, u_2 safer behaviour, u_3 testing, u_4 adherence.

Dashboard role: each slider changes the effective rates before running the simulation.

Fractional-Order Model

Caputo fractional dynamics

$$^{C}D_t^q X(t) = F(t, X(t)), \qquad 0 < q \le 1$$

$$^{C}D_t^q S = \Lambda - \beta_{\text{eff}} S(I + \eta T) - \mu S,$$
$$^{C}D_t^q I = \beta_{\text{eff}} S(I + \eta T) - (\tau_{\text{eff}} + \delta + \mu) I,$$
$$^{C}D_t^q T = \tau_{\text{eff}} I - (\rho_{\text{eff}} + \mu) T,$$
$$^{C}D_t^q A = \delta I + \rho_{\text{eff}} T - (d + \mu) A.$$

Intervention-adjusted rates

$$\beta_{\text{eff}} = \beta_0 (1 - u_1)(1 - u_2)$$

$$\tau_{\text{eff}} = \tau(1+u_3), \qquad \rho_{\text{eff}} = \rho(1-u_4)$$

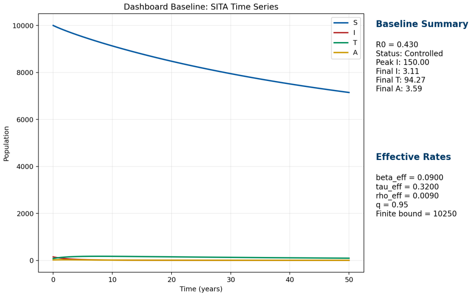
When $q = 1$, the model becomes ordinary. When $q < 1$, the model carries memory of past states.

Numerical Scheme + Dashboard

Computational implementation

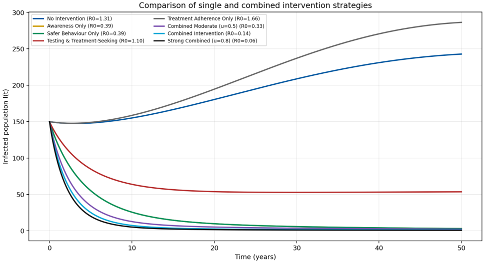
- ABM-type predictor-corrector solver for Caputo fractional equations.
- Real-time dashboard controls for parameters, initial conditions, intervention levels, and simulation horizon.
- Automatic outputs: trajectories, $\mathcal{R}_0$, scenario tables, memory comparison, sensitivity ranking, and exports.

Live system:
`https://hiv-model.onrender.com/`

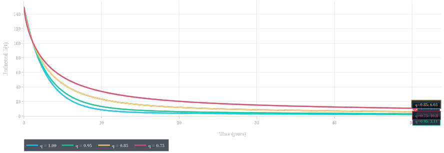


Dashboard baseline trajectory view.

Simulation Results



Combined interventions reduce the epidemic burden more strongly than isolated controls.



Changing q shows the visible effect of fractional memory on $I(t)$.

Interpretation: prevention, testing, treatment uptake, and adherence work best together; lower q values show that past behaviour and treatment history can affect long-term trajectories.

Live Demo Plan

Demo choreography

1. Open `https://hiv-model.onrender.com/`.
2. Show the moving SITA graph: $S(t)$, $I(t)$, $T(t)$, $A(t)$.
3. Switch from no intervention to combined intervention.
4. Compare $q = 1$ ordinary model with $q < 1$ fractional memory.
5. Open sensitivity and scenario tabs to defend the dashboard evidence.

What examiners should see

- The thesis equations are implemented.
- Graphs move because the model is being simulated.
- The dashboard produces chapter-ready figures and tables.
- The work is reproducible and deployed online.

Key phrase: "My main contribution is not only the model, but a working dashboard that allows the model to be explored live."

Conclusion + Recommendations

Conclusion

- A fractional-order SITA HIV model was formulated and implemented.
- Social behaviour interventions were incorporated through effective rates.
- The dashboard connects theory, computation, scenario analysis, memory effects, and sensitivity results.

Recommendations

- Continue integrating prevention and treatment-support interventions.
- Use Rwanda-specific data in future calibration studies.
- Extend the dashboard with uncertainty analysis and richer intervention scenarios.

Final message: the project transforms a fractional HIV model into an interactive Rwanda-aware simulation tool for academic analysis and communication.

Thank you

Live dashboard: `https://hiv-model.onrender.com/`